

SHAHADDIN GAFAROV

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EDUCATION

University of Florida

Ph.D. in Computer Science; Advisor: Dr. Tamer Kahveci

2025 – 2029

Gainesville, FL

University of South Florida

B.S. in Computer Science, Undergraduate Advisor: Dr. Yu Sun

2021 – 2024

Tampa, FL

EXPERIENCE

University of Florida

April 2025 – Present

Graduate Research and Teaching Assistant, Bioinformatics and Medical AI

Gainesville, FL

- Built the **first biological adversarial robustness benchmark** for dynamic graph-based link prediction algorithms, pitting 7 attack strategies against 6 victim models across 4 protein–protein interaction datasets, with 4 custom perturbation budgets.
- Designed a **deep learning model** for recovering unmeasured genes from **spatial transcriptomics** assays using a frozen single-cell foundation model.
- Worked on **SPACT**, a multi-modal cancer survival predictor fusing whole-slide histopathology with multi-omics data, accepted to the top computer-science-based medical imaging journal.
- Contributed to the design and developed the benchmarking suite of **PLATO**, a dynamic graph link prediction algorithm for complex biological networks.
- Served as **Lead Teaching Assistant** for “Advanced Data Structures” and “Enterprise Software Engineering Practices”.

University of South Florida

December 2023 – December 2024

Undergraduate Research Assistant

Tampa, FL

- Worked on **the detection of 6D poses** of top-layer objects for robotic grasping using deep learning.
- Developed benchmarking protocols for **multi-object robotic grasping**, contributing to a publication in IEEE Robotics and Automation Letters.

MusicLessonHub

April 2022 – August 2022

Full-Stack Developer

Boise, ID (Remote)

- Contributed to an early-stage **startup** connecting students and teachers across the United States.
- Built front-end interfaces and back-end endpoints for a student–teacher matching platform, working directly with designers to ship user-facing features.

AzeriMed LLC

June 2020 – June 2021

Software Engineer; Databases

Baku, Azerbaijan

- Built and maintained the database layer for aptekonline.az, Azerbaijan’s first online pharmaceutical platform, reducing query latency and error rate; the platform received the National Internet Award in 2023.
- Collaborated with developers on **database error handling**, backend reliability, and resolution of data-related issues.

PUBLICATIONS & MANUSCRIPTS

1. **Gafarov, S.**, Khatib, T., Yang, Z., Turan, M., Sun, J., Atay, Y., & Kahveci, T. *A Comprehensive Benchmark for Adversarial Robustness in Dynamic Biological Network Prediction*. Under review, ACM Computing Surveys.
2. Khatib, T., **Gafarov, S.**, Baktir, N., Gallardo, A. M., Atay, Y., & Kahveci, T. *Biological Network Prediction: A Survey of Dynamic and Static Algorithms*. Under review, Archives of Computational Methods in Engineering (Springer).
3. Ögülmüş, F. E., **Gafarov, S.**, Almaloğlu, Y., Özdemir, B. H., Ok Atılgan, A., Demir, D., Özen, Ö., Keleş, G. E., Kahveci, T., & Turan, M. (2026). *SPACT: A Clustering-Driven Multi-Modal Framework for Survival Prediction Using Genomic and Histopathology Data*. Medical Image Analysis, 112, 104078.
4. Khatib, T., **Gafarov, S.**, Strikowski, J. D., Turan, M., & Kahveci, T. (2025). *PLATO: Predicting Longitudinally-Aligned Time Observations of Biological Networks*. Proc. of the 16th ACM Int’l Conf. on Bioinformatics, Computational Biology and Health Informatics (BCB ’25).
5. Chen, T., Frumento, R., Pagnanelli, G., Cei, G., Keth, V., **Gafarov, S.**, Gong, J., Ye, Z., Baracca, M., D’Avella, S., Bianchi, M., & Sun, Y. (2025). *Benchmarking Multi-Object Grasping*. IEEE Robotics and Automation Letters, 10, 9510–9517.

TECHNICAL SKILLS

Languages	Python, C/C++, Java, SQL, JavaScript
Machine Learning	PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, Scanpy
Infrastructure	SLURM, HPC clusters, Git, Docker, CUDA

AWARDS

Bronze Medal	National Informatics Olympiad of Azerbaijan
Green & Gold Presidential Award	University of South Florida
Full-Ride Scholarship	University of Florida